

SOUMIL BHANDARI

703-843-4857 | soumil.bhandari@gmail.com | linkedin.com/in/soumilbhandari | github.com/soumilbhandari

EDUCATION

Purdue University, West Lafayette

Expected Dec 2028

B.S. Computer Engineering

BASIS Independent McLean

2026

High Honors Diploma | Capstones: Fourier Series, Linear Algebra, Multivariable Calculus | AP: Calculus BC, Computer Science A, Biology, Chemistry

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL, MATLAB, Bash, $\LaTeX$

Machine Learning: PyTorch, TensorFlow, HuggingFace, LoRA, LangChain, spaCy, scikit-learn, XGBoost, NumPy, Pandas

Tools & Frameworks: Git, Docker, AWS, Google Cloud, PostgreSQL/PostGIS, MongoDB, React, Node.js, Flask, Raspberry Pi, OpenSCAD

EXPERIENCE

AI Researcher, Perforated AI

Aug 2026 – Present

- Researching dendritic model compression

Research Intern, Harvard Medical School

Feb 2026 – Aug 2026

- Automated resting motor threshold (RMT) estimation, today an operator's by-eye call, with a Bayesian adaptive estimator fed by real-time ML classification of motor-evoked potentials
- Wrote the Python/MATLAB signal chain behind it (artifact rejection, feature extraction, MEP classification out of artifact-buried EMG), advised by Dr. Aapo Nummenmaa

Research Intern, Indian Institute of Technology (IIT) Roorkee

May 2025 – Feb 2026

- Built the classifier gating a clinical RAG system in a graduate AI/ML group: thousands of medical documents routed to the right corpus at 92%+ accuracy, cutting hallucination at the source
- Fine-tuned LLaMA-2 and Flan-T5 with LoRA on that corpus; automated retraining so new documents redeployed unattended

Software Engineering Intern, McKesson (No. 8 Fortune 500, \$359B revenue)

Jun 2025 – Aug 2025

- Cut analytics turnaround 35% by running 1M+ unstructured clinical records through a GPT-based extraction layer for iKnowMed, McKesson's oncology-specific EHR
- Wrote the Python/Java services integrating those AI workflows into iKnowMed and the surrounding enterprise EHR stack

PROJECTS

Neural Space Cap: Wearable Neuro-Cardiac Monitoring for Spaceflight

Sep 2026 – Present

- Purdue VIP team building an EEG and PPG cap that flags fatigue, cognitive stress, and early hypoxia in commercial spaceflight
- On the signal side: EEG denoising, artifact rejection, and LSL-synchronized multimodal capture

You2: A Voice Assistant That Remembers, Running Entirely On Its Own Hardware

Jun 2026 – Present

- Answers in under a second on a Raspberry Pi 5, storing nothing off-device, by overlapping transcription, generation, and speech
- Designed memory to consolidate rather than accumulate: a nightly pass summarizes, versions, and decays what it heard, so over a simulated year retrieval precision climbed 0.50 $\rightarrow$ 1.00 with superseded facts never resurfacing

Flipper: Real Estate Deal-Sourcing Engine (\$50K+ personally realized)

Apr 2026 – Aug 2026

- Scored Bright MLS listings on ARV, rehab, and spread via a PostGIS comp-radius model, with a Claude layer emitting max offer and risk flags
- Surfaced 500+ flip-ready properties and \$1.2M+ in identified spread across the DMV

Neurafy: Predicting Alzheimer's Progression (3 State Fair Awards, 3 Patent Filings)

Nov 2025 – Jun 2026

- Trained a multimodal progression model to AUC 0.95–0.97 in PyTorch, at or above SOTA published in *Nature* and far faster to train
- Founded it, self-funded \$10K+, signed a clinical pilot and added Dr. Benjamin Bolger (Harvard) as advisor; failed startup

AWARDS AND PATENTS

Named Inventor on 3 Provisional US Patents, Multimodal Neurodegenerative Modeling

Filed Dec 2025

- *Modality-Balanced Multi-Omic Modeling with Drift-Resilient Harmonization* (63/949,237)
- *Tri-Modal Network-Based Identification of Neurobiological Resilience Signatures* (63/949,231)
- *Quantifying and Predicting Cognitive Resilience Under Neurodegenerative and Stress-Induced Pathologies* (63/949,240)

Other Awards

2024 – 2026

- **Office of Naval Research, Yale Science & Engineering Association** (one per fair, for the top exhibit in CS, engineering, physics, or chemistry), and **Leidos** awards at the Virginia State Science & Engineering Fair, from a $\sim$ 200-exhibit state field across 10 regional fairs
- **Y Combinator AI Startup School:** 2,500 of 35,000+ applicants (top 4%) selected to learn from leading AI founders and researchers

Competitive Debate: 3 $\times$ Champion, Tournament of Champions Qualifier (top 0.5% nationally)

2024 – 2025

- **Champion, Columbia Invitational** (11–1): the nation's most competitive Public Forum field; top-ranked debater in school history
- **Champion, WACFL Public Forum 4 and 5** (4–0 each): undefeated at the largest regional tournament to date, over the top DC-area teams
- **Quarterfinalist, George Mason** (1st seed off undefeated prelims) and **Octofinalist, Sunvite** (5–1); 1st seed at Coolidge